# SANKALP

**Robotics Mechanical Design Engineer**

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## SUMMARY

Mechanical design engineer building end-effectors and actuation mechanisms for autonomous construction robots and cobots at Origin, from concept CAD and FEA/CFD through DFMA, sheet-metal design and prototype testing. 2 published patents, 1 ASME paper; **AIR 1, Autodesk Design Challenge** and **Smart India Hackathon 2024 winner**.

## EXPERIENCE

**Origin — Autonomous Construction Robotics** **Bengaluru | May 2025 – Present**

*Robotics Mechanical Design Engineer (Jul 2026 – Present) · Intern (May 2025 – Jul 2026), PPO Dec 2025*

- Designed **swappable sander and sprayer end-effectors** in Onshape, letting one robot switch finishing tasks.
- Designed an **offset sander end-effector for a UR12e cobot** with **servoing and vision capabilities** for robotic surface finishing.
- Engineered a **pneumatic-cylinder-actuated bucket-swap mechanism** for fast tool and material changeover on the robot.
- Re-designed structural and tooling parts for **sheet-metal fabrication using DFMA** to simplify manufacturing and assembly.
- Received a **Pre-Placement Offer (PPO) in Dec 2025** (7th sem.) for owning tooling sub-systems; full-time from Jul 2026.

**Adelix Pvt. Ltd. — Design-for-Manufacturing Intern** **Aug 2024 – Oct 2024**

- Designed a **universal docking system for medium-class multirotor UAVs**, including a gear- and drag-based cable uncoiling/recoiling mechanism driven by integrated motor drives.

**Anusandhan National Research Foundation (DST, Govt. of India) — Summer Research Intern** **Jun 2024**

- Designed a tabletop fume-hood ventilation system; optimised airflow via CFD and structure via topology optimisation.

## PROJECTS

**Autonomous Multirotor UAV — SAE Aerothon 2024** [Report] | *SolidWorks, Ansys, LS-Dyna, ArduPilot* **May 2024 – Jul 2024**

- Captained Team Air Guardians: 2 kg quad (**2.54:1 T/W**, **15 min**), YOLOv4-Tiny vision; won **Best Autonomous Mission**.
- Validated the chassis via Ansys modal analysis (first natural frequency 252 Hz) and an LS-Dyna 30 m drop simulation (**FoS 2.36**); redesigned the stack cover to cut drag **21%** (2.08 → 1.65 N).

**eVTOL Quad-plane for Medical Logistics** [Report] | *Fusion 360, Autodesk CFD, ROS 2, Gazebo, URDF* **Mar 2024 – Dec 2024**

- Sized a **40 kg-payload**, 50+ min VTOL with CFD-validated morphing wing; **built & flight-tested an SIH 2024 prototype**.
- Built a full-vehicle URDF from CAD; ran controller-in-the-loop, payload-drop and waypoint tests in **ROS 2 + Gazebo**.

**Fixed-Wing UAV — SAE ISS DDC 2024** [Report] | *XFLR5, Ansys Fluent, SolidWorks* **Feb 2024**

- Chose S1223 airfoil from 6 candidates ($C_L$ 1.18 vs ≤0.5 at 0° AoA) and a 30%-tapered 1.25 m wing (+15% lift); built a laser-cut balsa/Depron airframe; won **2nd, Best CFD Analysis**.

**Extrusion-Based Ceramic 3D Printer ($B_4C$)** [Deck] | *SolidWorks, Ansys Fluent, Arduino/C++* **Feb 2024 – May 2024**

- Retrofitted a Creality Ender 3 with a syringe-pump extruder (NEMA 23, TB6600, lead screw) and **closed-loop Arduino control** with ultrasonic feedstock sensing and alerts; led to a **patent application and ASME paper**.
- Characterised 5 slurry compositions and validated nozzle flow in Ansys Fluent at 10% and 100% feed rates.

## PATENTS & PUBLICATIONS [Patent filings]

- **Patent:** An Extrusion System for 3D Printing and a Method of Operation Thereof — IN 202521020630
- **Patent:** Electric (eVTOL) Aircraft for Medical Logistics and Disaster Surveillance — IN 202521057108; design **registered**
- **Paper:** Evaluating Extrusion Dynamics in Ceramic 3D Printing Using FEA — ASME IMECE India 2025

## AWARDS & CERTIFICATIONS [Certificates]

- **AIR 1**, Autodesk Design Challenge 2024 (IIT Kanpur) · **Winner**, Smart India Hackathon 2024 (Hardware) · **Best Autonomous Mission**, SAE Aerothon 2024 · **AIR 2**, Best Aerodynamic Analysis (CFD), SAE ISS Drone Development Challenge 2024
- Certified SolidWorks Associate (CSWA) · Top 5% nationwide, Aircraft Design (NPTEL, IIT Kanpur) [Portfolio]

## SKILLS

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|---|---|
| **CAD & Design** | Onshape, SolidWorks (CSWA), Fusion 360; sheet-metal, DFMA, mechanism design, drawings |
| **Simulation** | Ansys Structural/Modal/Fluent, LS-Dyna, Autodesk CFD, XFLR5; FEA, CFD, topology optimisation |
| **Robotics & Software** | Python, C++, OpenCV, ROS 2, Gazebo, URDF, ArduPilot, Pixhawk, Arduino |
| **Manufacturing** | Additive manufacturing (FDM, ceramic extrusion), sheet-metal fabrication, prototyping |

## EDUCATION

**Indian Institute of Information Technology, Design and Manufacturing (IIITDM), Jabalpur** **2022 – 2026**

*Bachelor of Technology (B.Tech), Mechanical Engineering* *CPI: 8.6 / 10*